777 Radio: First Pipeline — Draft Proposal

CS156 | Pre-class draft | October 8, 2026

Status: Research plan and exploratory dataset inspection. Audio extraction, model training, and quantitative evaluation are proposed work, not completed results.

1. Project motivation and question

I built 777 Radio, an image-to-song recommendation app. The current system uses a generative AI API to describe an uploaded image and matches its aesthetics to tags that I manually assigned to songs. Hand-tagging every new track limits how much I can expand the library.

Primary question: Can a machine-learning model learn my detailed aesthetic song tags from actual audio and automatically tag unseen songs well enough to preserve the quality of image-to-song recommendations?

Secondary extension: Can a local pretrained image encoder replace the paid image-description API without significantly reducing recommendation quality? This is optional for the first assignment.

2. Personal dataset and existing code

The existing **public/music-dataset.csv** contains **162 songs** with columns **song_id, title, artist, context_tags, spotify_id**. Every song has manually curated comma-separated tags, often spanning moods, colors, settings, fashion, and cultural references. 160 Spotify IDs are missing. The archive also includes **app/api/analyze-image/route.ts**, which implements image analysis and song ranking.

Important data-quality issue: The current app assigns random placeholder values to energy, valence, and tempo. These are not acoustic measurements and must not be used as training features. The CSV also does not contain audio recordings. Actual audio access is a prerequisite for sound-based ML.

3. Proposed pipeline

Existing song CSV + legally accessible song audio → clean and standardize tags → extract acoustic features (e.g., tempo, RMS energy, spectral centroid, MFCC summaries) → split by song into training/test sets → train multi-label models → predict tags for held-out songs → compare with human tags → optionally feed predicted tags into 777 Radio and evaluate image-song recommendations.

4. Three comparison conditions

Condition	Approach	Purpose
Baseline	Predict common training-set tags	Check whether ML beats a trivial strategy
Model 1	One-vs-rest logistic regression	Simple class model on acoustic features
Model 2	One-vs-rest SVM	Test a more flexible classifier

5. Evaluation and potential problems

Use a held-out test set split **by song** (approximately 80/20) and compare multi-label precision, recall, F1, and top-k tag recovery. Because tags are subjective and many are rare, also inspect per-tag frequencies and errors. Avoid using existing human tags as both input features and target labels: that would leak the answers. If audio is split into clips, keep all clips of one song in the same partition.

Working app: https://777radio.vercel.app/

A secondary image-matching evaluation could use independently labeled image-song pairs and compare the original human-tag recommendations with automatically tagged songs using Precision@3. This requires collecting image-song relevance judgments, and is not yet completed.

6. Connection to first-unit coursework

This plan builds on the forum workbook's train/test split, logistic regression, SVM, and model evaluation. It also connects to Naive Bayes, decision trees, overfitting, and generalization discussed in earlier CS156 classes. Unlike the workbook's simple binary circle data, my task has multiple labels per song.

7. Immediate next steps

Audit the tag vocabulary and class frequencies; confirm lawful access to audio files; extract measured acoustic features; establish a reproducible train/test split; run baseline, logistic regression, and SVM; plot results; test a small image-song matching set; document limitations. The attached starter notebook already performs the CSV audit and prepares a song-level split.

Sources: my 777 Radio source archive (vibe-match-app-3.zip); Generated Forum Code Workbook (course resource); scikit-learn documentation (implementation reference to consult).